

Math 103

Class 6

2. Mean, Median, Mode

3. Mean (average) $\bar{x} = \frac{\sum x}{n}$ $\Sigma = \text{sigma}$
"Find the sum"

Data: 5, 10, 7, 14, 16, 14

$$\bar{x} = \frac{\sum x}{n} =$$

The mean, in most cases, is not an actual data value.
Round to one more place than the raw data.

4. Median

Midpoint of a data array.

Data: 5, 10, 8, 7, 9, 8, 6

Arrange in order: 5, 6, 7, 8, 8, 9, 10

Median = 8

If there are an even number of data values:

Data: 22, 21, 25, 28, 25, 21, 37, 22, 26, 20

Arrange in order: 20, 21, 21, 22, 22, 25, 25, 26, 28, 37

The median is located halfway between 22 and 25.

$$\text{Median} = \frac{22+25}{2} = 23.5$$

5. Mode (The value that occurs the most often)

Data: 4, 10, 8, 7, 7, 6, 5, 3

Mode = 7

Data: 4, 9, 12, 21, 6, 3, 8

No mode

Data: 5, 6, 6, 7, 6, 3, 4, 5, 5, 9

Modes = 5, 6 (bimodal)

(When there is a big discrepancy between the median
and the mean, then there are some "outliers".)

6. Midrange

$$\frac{\text{Lowest} + \text{Highest}}{2}$$

7. More practice:

146	152	168	174	180	178	179
180	178	178	168	165	152	144

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Answers:  mean = 167.3
          median = 171
          mode = 178
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23 17 35 30 26 39 45 39 13 42 285 394

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Answers:  mean = 82.3
          median = 37
          mode = 39
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89	50	68	60	375	55	500	71	40	350
60	50	250	45	45	125	235	65	60	130

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Answers:  mean = 136.2
          median = 66.5
          mode = 60
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H.W: Find the mean, median, mode, midrange and variance

Find the weighted average:

30% HW - 100 80 80 80 100 70 70 70 70 (80)

50% Tests - 85 89 93 70 (84)

20% Final Exam - 80 (80)

Average (82)